

Valuing off-ball runs with StatsBomb 360

A possession-value model for the movement that happens away from the ball, and a player metric built on top of it.

Pranav Mohan · Data Scientist assessment, Parts 1 & 2 · July 2026 · Code: github.com/pranavm28/statsbomb-off-ball

Headline. Event data records the pass; it never records the run that made the pass possible. Using 360 freeze-frames I reconstruct 61,823 off-ball runs and value each one by how much the *manner* of the movement raises the chance the possession progresses. Seeing the players — not just the ball — is worth +0.0212 AUC over event-only features on identical rows and folds. The applied finding is a profile, not just a ranking: **Victor Boniface** ranks second per 90 on a fifth of Mbappé's minutes, while receiving in the most crowded space of anyone near the top.

1 What "possession value" means here

Possession value here is a number attached to a *state* of the possession — not to a player and not to an action, but to a moment: the ball is here, this player is there, the defenders are arranged like this. The value of a move is then the *change* in that number between two states. That logic is shared with xT and VAEP; what differs is what goes into the state.

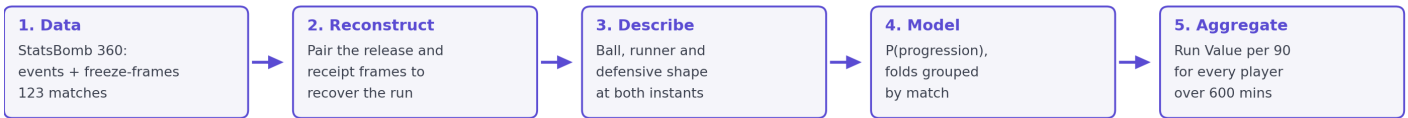
Two models do two different jobs. They are kept separate on purpose, and knowing which is which is needed to read every number below.

	Model A — state value	Model B — run value
Question it answers	Does seeing the <i>players</i> , not just the ball, make a game state more predictable at all?	How much did <i>this</i> player's movement add, over and above where his run started and finished?
One row is	one game state: a player and the ball at one instant	one reconstructed off-ball run
Rows	185,967	61,678
Target — the 0/1 label it learns from	a shot within the next 5 seconds	progression: a shot or a final-third entry within the next 5 actions of the possession
How often the label is 1	3.3%	33.9%
What it is used for in this document	the 360 ablation in §6 — does 360 earn its keep	Run Value per 90 — the metric and ranking in §4

Why two targets instead of one? "Shot within five seconds" is right for the first question: it is close to what football cares about, and across 185,967 states there is enough data to learn an event that rare (3.3%). It is wrong for the second. At the level of one run, shots are too rare to tell a good run from a lucky one, so Model B is judged on progression, which happens 33.9% of the time. A run that drags the ball into the final third has done its job whether or not a shot arrives three passes later.

Why a learned state-value function rather than a grid xT? xT keys value to *where the ball is*; this keys it to *where the players are*. Two identical ball positions with different defensive shapes are one cell in xT and two distinct states here. An xT surface is still used, as a context *feature*, so the model knows the geographic prior it is adding to — the threat surface is an input, never the thing being predicted.

2 How it works



The reconstruction does the real work. A **Ball Receipt** event carries its own freeze-frame with the receiver tagged **actor**, which fixes where the run *ended* and who made it. The preceding pass's frame fixes where everyone stood when the ball was struck — but frames are anonymous, so the run's *origin* must be inferred: the nearest teammate in the release frame, accepted only if the implied sprint is physically possible (≤ 9.5 m/s over the pass duration). This origin inference is the biggest assumption in the project.

Two filters, in order. Of 93,202 receipts examined, 84% pass the plausibility gates — an unambiguous nearest teammate and a reachable distance. Of those, the ones where the player actually *moved* (≥ 2.0 m) are runs: 61,823, or 66% of all receipts. The rest are players receiving roughly where they already stood, which is a reception, not a run.

3 Scope, and why

Four full competition-seasons with 360 coverage — La Liga 2020/21, Ligue 1 2021/22 and 2022/23, Bundesliga 2023/24: 123 matches, 264,814 player-minutes. I pooled seasons rather than use the World Cup because off-ball running needs volume per player and the metric is a rate: a tournament gives seven matches, a season thirty-plus. The honest cost is that open-data 360 covers one focus club per competition, so the pool is effectively three clubs and the ranking is within-sample, not cross-league.

4 The metric, and the ranking

Run Value per 90 is built from **Model B**, so the quantity underneath it is the probability of *progression*, not of a shot. Four steps:

- (i) **Fit Model B twice on identical rows and folds.** Once on CONTEXT only — `origin_x/y`, `receipt_x/y`, threat at both ends, pass length: *where* the run happened. Then again on CONTEXT + RUN — distance, forward and lateral components, speed, separation gained, space at receipt, defenders broken, in-behind, encirclement and block geometry: *how* he moved.
- (ii) **Subtract, per run.** `run_value_added = P(progress | where + how) - P(progress | where)`. Both are out-of-fold predictions, meaning each run is scored by a model that never saw that run's match in training. This difference is the run's credit.
- (iii) **Keep only runs received in the final third.** Outside it, "predictive" and "valuable" come apart: a centre-back dropping into space reliably predicts retained possession and creates nothing. Tested against the alternatives — the attacking half moved the clearest false positive from

10th to 15th, the final third moves him to 34th.

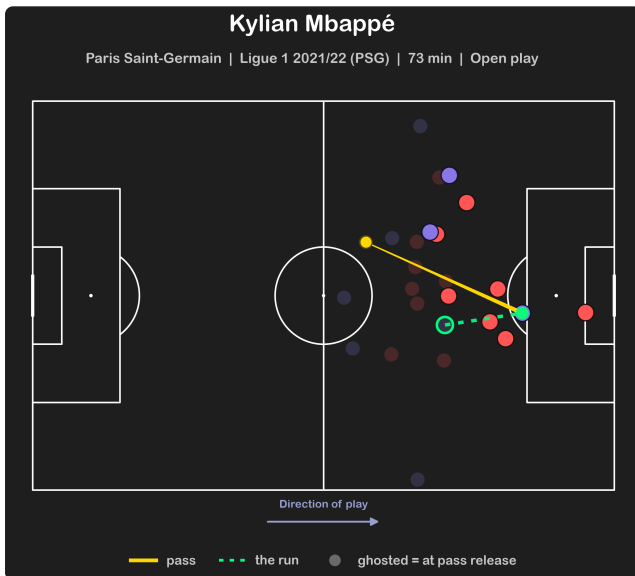
- (iv) **Sum per player, divide by 90s played**, with a 600-minute floor. That leaves 59 qualifying players.

It is an information gain, not a probability. It answers "how much of the outcome is explained by the manner of the movement rather than its geography", so it does not sum to goals. Two scouts watch the same possession, one seeing only the ball, one seeing the runner: Run Value is what the second knew that the first did not.

#	Player	Team	Position	90s	Runs /90	F3 runs /90	In behind /90	Enclosed share	Run Value /90
1	K. Mbappé	Paris Saint-Germain	Forward	50.6	23.3	13.7	2.03	0.40	0.616
2	V. Boniface	Bayer Leverkusen	Forward	17.8	13.2	7.5	1.23	0.52	0.590
3	F. Wirtz	Bayer Leverkusen	Left Att Mid	27.3	29.7	13.0	1.39	0.44	0.563
4	F. Trincão	Barcelona	Right Wing	8.2	26.8	15.7	1.71	0.30	0.552
5	M. Braithwaite	Barcelona	Forward	11.8	12.2	8.0	1.11	0.46	0.519
6	J. Hofmann	Bayer Leverkusen	Right Att Mid	25.2	25.3	10.8	0.75	0.40	0.499
7	Neymar	Paris Saint-Germain	Left Wing	31.7	33.2	13.5	1.55	0.40	0.469
8	P. Schick	Bayer Leverkusen	Forward	12.5	9.8	6.2	0.80	0.59	0.459
9	H. Ekitike	Paris Saint-Germain	Forward	11.4	12.5	6.9	0.79	0.52	0.454
10	N. Tella	Bayer Leverkusen	RWB	10.1	11.2	6.4	1.09	0.19	0.343

Enclosed share = mean angular encirclement on receipt (1 = defenders on all sides, 0 = a completely clear side). F3 = final third.

5 One run, in detail



Ghosted dots = positions at pass release; solid = at arrival. Gold is the ball, green dotted is the run.

Mbappé starts **outside the box** at the moment of release and arrives on the penalty spot 16.2 m later, with **four defenders** around him — encirclement 0.63, i.e. almost no clear side. The possession produces **0.70 xG within five seconds**.

Scored, step by step. From where the run started and finished alone, Model B put the chance of progression at **0.774**. Once it could also see *how* he moved — 16 m forward, in behind, into encirclement 0.63 — it rose to **0.933**. The difference, **+0.159**, is this run's credit. The possession did progress.

That gap is the whole idea: without the frames this is just a 16 m pass into a crowd.

6 Validation

Nested structure. Events sit inside possessions inside matches, so a random split leaks — two events from one possession are near-duplicates and would land on both sides of the split. All folds are `GroupKFold` on `match_id`, so a whole match sits either in training or in test, never both. Every number quoted is out-of-fold: predicted by a model that never saw that match.

Does 360 earn its keep? (Model A.) The honest test is not "with vs without 360" — the runner's own position is a 360 feature, so a naive baseline smuggles it in. That mistake is worth +0.0069 AUC of flattery. Instead, four nested tiers on identical rows and folds:

Feature tier	n	AUC	Brier	Gain	What it adds
1. Event data only	3	0.9238	0.0260	—	what event data alone can see
2. + where the runner is	7	0.9375	0.0244	+0.0137	the runner's own position, from 360
3. + defensive structure	12	0.9444	0.0238	+0.0069	where the defenders are
4. + compactness / encirclement	16	0.9450	0.0238	+0.0006	shape of the block (negligible)

Tier 1→3 is the real 360 contribution: **+0.0212 AUC**. Tier 4 is **+0.0006** — a null result, reported rather than dropped. Read the *increments*, not the levels: an AUC of 0.92 on tier 1 looks impressive but the target is easy, since a ball already in the box implies a shot soon. Only the incremental columns say anything.

Does the movement earn its keep? (Model B.) The same design, one level up: CONTEXT alone reaches AUC 0.7806; adding the RUN features reaches 0.8002, a lift of **+0.0197** on 61,678 runs. That lift is what Run Value is a per-run decomposition of. **Calibration:** out-of-fold Brier 0.1630 for Model B against a 33.9% base rate, and 0.0238 for Model A against 3.3%; predicted versus observed track closely across all ten deciles for both (figures in the repo).

7 Say it two ways

To a sporting director (~200 words)

We can now put a number on something we could previously only describe: what a player's running *without* the ball is worth.

For every pass in our sample we can see where all twenty-two players stood when the ball was struck, and again when it arrived. That lets us rebuild the run the receiver made to get there and ask one question: how much more likely did a shot become because of the way he moved, rather than simply because of where the ball ended up?

Mbappé finishes top, which tells you the method isn't broken but doesn't tell you anything you'd pay for. The name worth your time is **Victor Boniface**. He is second on this list on roughly a fifth of Mbappé's minutes, and he does it while receiving in the most crowded areas of anyone near the top — more than half his receptions come with defenders closing him from most sides. That is a specific and buyable habit: a forward who will still move into a packed box, and whose movement makes the pass playable.

How much would I bet? Enough to put him on a shortlist and watch him properly. Not enough to price a bid on this alone — the sample is three clubs, and below the leading names the order shifts depending on which outcome we ask about.

To the data team (2–3 sentences)

Run Value is an ablation quantity: the out-of-fold gain in $P(\text{shot} \leq 5s)$ once the model can see *how* the player moved, on top of a context model that already has ball location, receipt location and defensive structure — so it is information gain about the manner of movement, not a probability, and it will not sum to goals.

Folds are `GroupKFold` on `match_id`; the 360 tiers are worth +0.0212 AUC over event-only features on identical rows and folds, and out-of-fold predictions are calibrated (Brier 0.0238 on a 3.3% base rate).

What would make me trust it more: a counterfactual attribution that holds the ball and the defenders fixed and moves only the runner, because the current quantity is target-sensitive — three of four target definitions I tested reorder the leaderboard, which is a property of information-gain metrics rather than a bug, but it needs pinning down before anyone scouts off it.

8 What I used, and what I rejected

Kept

- **Ball Receipt freeze-frames.** The receiver is tagged `actor` — the only route to the runner's identity.
- **Nearest-teammate origin with a speed gate.** Frames are anonymous; the gate keeps physically impossible matches out.
- **LightGBM.** Tabular, non-linear, handles the mixed feature scales; the relationships here (distance, angle, congestion) are not linear.
- **Angular encirclement** over defender counts — *where* the defenders are beats *how many*.
- **Final-third restriction** to stop rewarding informative-but-harmless movement.

Rejected, and why

- **socceraction / VAEP off the shelf.** The point was defining the target myself; a black box is the failure mode the brief names.
- **Defender counts within 10 m.** Collinear with the geometry, no lift.
- **Compactness tier.** +0.0006 AUC — kept in the report as a null, not in the metric.
- **Plain ΔV as the player metric.** It is joint with the pass; $\text{corr}(\Delta V, \text{ball forward})$ exceeded $\text{corr}(\Delta V, \text{run forward})$, so it partly ranked passers.
- **Pitch control / velocity models.** Freeze-frames have positions, no velocities.
- **World Cup 2022.** Too few matches per player for a rate metric.

9 What breaks it

The one that matters. The metric is **target-sensitive**: of four target definitions tested, three reorder the leaderboard and two produce football-nonsense orderings (defenders and deep midfielders on top). That is inherent to an information-gain quantity — changing the outcome changes what counts as position-explained versus movement-explained — but the ordering below the leading names is not settled. The specified-but-unbuilt fix: a **counterfactual attribution** holding ball and defenders fixed and moving only the player, isolating his contribution instead of inferring it from a model comparison.

- **Who it treats unfairly.** Target men and deep-lying creators. It rewards movement that moves *this* possession forward quickly, so a striker who occupies two centre-backs to open space for someone else scores nothing for it. Fixing that needs off-ball value attributed to the *space created for teammates*, not just the receiving player.
- **Small samples.** The 600-minute floor is $\sim 6.7 \times 90$, so F. Trincão ranks 4th on 8.2×90 . Bootstrap intervals per player are the fix; they are not in this version.
- **Three clubs.** Open-data 360 is club-focused, so this is within-sample.
- **Inferred origins.** 34% of receipts are discarded, and the kept ones rest on a nearest-teammate assumption that will occasionally pick the wrong player in a crowded box.

10 AI usage

The full log is `AI_USAGE.md` in the repo: which tools, where they were used, representative prompts, and for each substantive step what I accepted, what I modified and what I rejected. The short version: the framing and the metric concept are mine — model off-ball runs as a supervised, xG-style problem rather than a grid. The assistant recommended LightGBM with `GroupKFold` and drafted the defensive geometry; I rejected the density counters and required the ablation before believing any of it. The two interventions that changed the result were both verification, not generation:

- **A mislabelled baseline.** The first "without 360" comparison contained a 360 feature — the runner's own position — which flattered the result by +0.0069 AUC. Replacing it with the four-tier ablation gave the real figure, **+0.0212**.
- **A feature read backwards.** Block spread was about to be reported as "a stretched defence is more valuable". Checked against the data, it says the opposite by $7.6\times$ — it is a depth proxy, not a width one. It stayed in the model and out of the story.